

# 1 Explainability in Reinforcement Learning

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# Chapter 1

## Latent Space Analysis

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep Learning (DL) introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, Neural Network (NN) gradually construct representations that become increasingly relevant to the task under consideration [Rumelhart et al., 1986]. These representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training. Since these representations are learned internally by the network and are not directly observable from the input or output spaces, they are commonly referred to as Latent Representation (LR). The collection of these LR learned throughout the network is commonly referred to as a Latent Space (LS).

These LS are now at the core of a growing number of practical applications. Neural video compression exploits the compactness of LS to encode information efficiently [Lu et al., 2019]. In Reinforcement Learning (RL), agent internal representations are used for efficient exploration in extremely open environments [Burda et al., 2018]. Recommendation systems model users and content within a shared LS to compute meaningful similarities [He et al., 2017]. Retrieval-augmented generation systems index and query knowledge bases through latent embeddings [Lewis et al., 2021]. Finally, Large Language Model (LLM) and multimodal architectures rely on the quality of learned representations to align heterogeneous modalities such as text, images, and audio [Radford et al., 2021]. These examples represent only some of the most widespread applications of LR.

Given their extensive use, the study of LR has become a central topic in DL research. A major objective is to understand how information is organized within these spaces, which concepts are encoded, and how these representations

influence model behavior.

- Section 1.1 introduces the working hypothesis that underlies much of the research on LS analysis, which we refer to as the *Concept Hypothesis* (Hyp. 1).
- Sections 1.2, 1.3, 1.4, and 1.5 examine different approaches to LS analysis, respectively covering probing methods, semantic directions, sparse autoencoders, and clustering-based methods.
- Section 1.6 concludes with a comparative synthesis of these methods and a discussion of the open questions that remain.

## 1.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on LS analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, LS, concept, and semantics. Throughout this manuscript, we refer to this hypothesis as *Concept Hypothesis* (Hyp. 1). The goal of this section is to make this conceptual framework explicit by defining its main components.

The *Concept Hypothesis* (Hyp. 1) originates from the need to explain the generalization ability of deep NN observed after training, as discussed in Section ?? . A model is said to generalize when, for an input  $x \in \mathcal{X}$  not seen during training, it still produces an output  $\hat{y}$  close to the true output  $y$ . Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task [Bengio et al., 2014].

### Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction.

The ability of NN to perform abstraction arises from their intermediate activations. A NN does not directly map  $x$  to  $y$ , it computes a sequence of intermediate activations across hidden layers. These activations are numerical encodings of the input referred to as embeddings. Once training is complete, these embeddings are structured to support the performance of a specific task. To mark this distinction, we will refer to these trained embeddings as LR, as defined in Definition 1.1.2.

### Definition 1.1.2: Latent Representation

A representation is the internal encoding of an input given by a intermediate layer of a trained NN.

Each hidden layer refines the representation of the previous layer. Each transformation progressively reshapes the representation into a form more directly

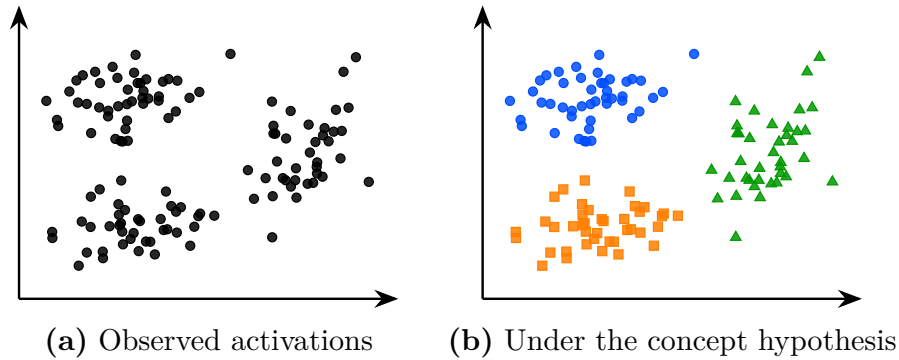


Figure 1.1: The same latent space activations seen without (a) and under (b) the *Concept Hypothesis* (Hyp. 1).

aligned with the output space [Rumelhart et al., 1986]. This process can be described as a sequence of transformations where each  $h^{(l)}$  is a representation:

$$x \rightarrow h^{(1)} \rightarrow h^{(2)} \rightarrow \dots \rightarrow h^{(L-1)} \rightarrow \hat{y}$$

The set of all such representations produced by a layer over a dataset defines its LS, as defined in Definition 1.1.3.

**Definition 1.1.3: Latent Space**

The LS is the set of all representations produced by a layer over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

Under the *Concept Hypothesis* (Hyp. 1), the analysis of the LS is assumed to provide access to abstract structures learned by the model, as illustrated in Figure 1.1. The same set of latent activations can be viewed either as an unstructured point cloud or, under this hypothesis, as a collection of points organized by concept membership, in which case the structure of the LS becomes meaningful and amenable to analysis. In Figure 1.1, the colors represent hypothetical concept memberships assigned by the model rather than observed labels. These structures are referred to as concepts. In this manuscript, a concept is defined as an abstract representation that organizes information and supports reasoning (see Definition 1.1.4).

**Definition 1.1.4: Concept**

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

Our definition of concepts is consistent with both the pragmatic tradition in philosophy and the information-processing view of cognition. Within these frameworks, concepts arise from the need to simplify an otherwise intractable reality. Because the environment contains more information than any cognitive system can process, intelligent behavior relies on abstractions, called concepts, that discard irrelevant details while preserving the information required to achieve

a given objective [Peirce, 1878, Misak, 2013]. These concepts are realized as internal representations that reduce complexity and enable reasoning under limited computational resources [Newell, 1980, Simon, 1996]. Although these theories were originally developed to explain human cognition, the *Concept Hypothesis* (Hyp. 1) assumes that the same principles can be extended to deep NN.

In order to influence the model’s computations, concepts must necessarily admit a concrete realization within the LS. We refer to this realization as the Concept Structure (CS). The definition of the CS is provided in Definition 1.1.5.

#### Definition 1.1.5: Concept Structure

A CS is the specific form for how a concept is instantiated within the LS.

The methods reviewed in this chapter all build on the *Concept Hypothesis* (Hyp. 1) formalized in Hypothesis 1, but differ in how they characterize the CS of a concept. This choice directly determines how concepts can be identified and extracted from the LS. As a result, several families of LS analysis methods have emerged in the literature, each adopting a different view of how concepts are structured within the LS.

#### Hypothesis 1: Concept Hypothesis

NN generalize by performing abstraction. Through training, their LS becomes organized around concepts. Concepts are task-dependent abstractions that capture regularities relevant to the task. Analyzing the LS therefore provides access to the CS through which these concepts are instantiated in the model.

Now that the definition of a *Concept Hypothesis* (Hyp. 1) has been established, we can consider its relationship to human. According to our definition, we can imagine concepts that are useful for the machine but do not make sense to humans. This relationship is captured by the notion of semantics [Marconato et al., 2023], defined in Definition 1.1.6. Some concepts developed in LS can be readily associated with human concepts and are said to have high semantic value. Others may contribute to task performance while remaining difficult for humans to understand, and are said to have low semantic value.

#### Definition 1.1.6: Semantics

Semantics refers to the alignment between concepts in the model LS and concepts used by a human observer. A concept is semantic if it can be associated with a human interpretable concept.

It is important to note that this conceptual framework is not supported by a complete theoretical understanding of deep NN. Current mathematical tools do not provide a rigorous characterization of their internal representations and do not formally prove that concepts emerge as identifiable CS within LS. Nevertheless, this perspective constitutes a widely adopted working hypothesis in the field of representation and LS analysis.

The remainder of this chapter is devoted to reviewing methods for analyzing LS and extracting the concepts they encode. We begin with probing, one of the

most direct approaches for assessing whether concepts are represented in the LS.

## 1.2 Probing

Probing methods aim to determine whether a given semantic concept is encoded within the LS of a NN [Li et al., 2024, Karvonen, 2024]. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The assumption is that if a concept can be reliably decoded from these representations, then the network likely encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of semantic concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concepts. We denote by  $c_j$  the presence or absence of concept  $j$  in example  $x$ . The dataset can therefore be written as follows:

$$\mathcal{D} = \{(x^{(i)}, y^{(i)}, c_1^{(i)}, \dots, c_n^{(i)})\}_{i=1}^N.$$

The data are subsequently projected into the LS of the pretrained model. We denote by  $h^l$  the representation vector obtained at layer  $l$ . From these LR, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe and is denoted by  $p$ . Its purpose is to determine whether a given concept is present in the representation. Its purpose is therefore to assess whether information associated with the concept is accessible from the network’s internal representations. This can be formalized as follows:

$$p_{\theta}^{l,j} : h^l \mapsto c_j, \quad \theta^* = \arg \min_{\theta} \mathcal{L}(p_{\theta}^{l,j}(h^l), c_j)$$

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to information explicitly encoded in the LS. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers.

A linear separation suggests that the information is easily accessible to the NN and potentially usable during decision-making. If the classifier achieves performance significantly above chance level, this suggests that the LS contains exploitable information related to the considered concept. Figure 1.2 illustrates the decision boundaries of such linear probes for three concepts, where each colored region indicates the area classified positively by the corresponding probe.

An application of this technique can be found in studies investigating whether LLM develop internal representations of their environment [Li et al., 2024, Karvonen, 2024]. In these works, researchers train LLM architecture to play games such as chess or Othello in a purely textual setting. The model is provided only with sequences of moves and information about the actions it can take, all expressed in textual form. The central question is whether the LLM develops an internal representation of

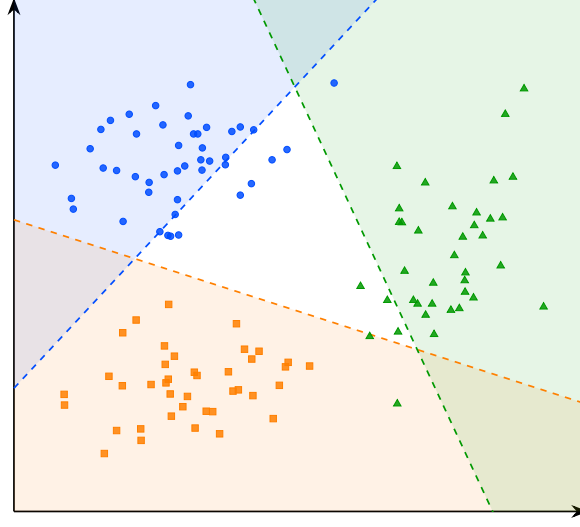


Figure 1.2: Decision boundaries of the linear probes for concepts A, B, and C.

167 the game state. To address this question, the probed concepts correspond to the  
 168 presence or absence of different game pieces on specific board positions.

169 As illustrated in Figure 1.3, probes are able to reconstruct the full board  
 170 state from the model’s hidden representations, even though the only available  
 171 information to the model is a sequence of text. In this figure, each row corre-  
 172 sponds to a piece type, comparing the ground truth with the probe’s predictions  
 173 and confidence. The close agreement between the ground truth and predicted  
 174 positions suggests that the model develops a structured representation of the  
 175 underlying environment from which the data are generated.

176 However, the fact that a concept can be decoded from LR does not necessarily  
 177 imply that the model actually uses this information to perform its task. To assess  
 178 whether representations play a causal role in the model’s output, it is possible  
 179 to perform interventions on the LR. An intervention consists of modifying a  
 180 LR in a controlled manner by activating or deactivating a given concept, and  
 181 then studying its impact on the model’s output. If the resulting change in the  
 182 model’s output is consistent with the applied perturbation, this suggests that  
 183 the representation had a causal role.

184 To this end, the LR  $h^l$  is minimally modified such that the probe prediction for  
 185 the target concept  $c_j$  changes state, while the predictions associated with other  
 186 concepts remain unchanged. This goal can be formulated as an optimization  
 187 problem over the latent activation itself. The optimization is performed directly  
 188 on the representation  $h^l$ , using gradient-based optimization as introduced in  
 189 Section ???. The solution to this problem is the optimal modified representation  
 190  $h^{l*}$ , defined as:

$$h^{l*} = \arg \min_{h^{l'}} \underbrace{\gamma \mathcal{L}_{\text{flip}}(p_{\theta}^{l,j}(h^{l'}), c_j)}_{\text{flip target concept } c_j} + \underbrace{\lambda \sum_{k \neq j} \mathcal{L}_{\text{preserve}}(p_{\theta}^{l,k}(h^{l'}), c_k)}_{\text{preserve other concepts } c_{k \neq j}} + \underbrace{\beta \|h^{l'} - h^l\|_2^2}_{\text{minimality}}$$

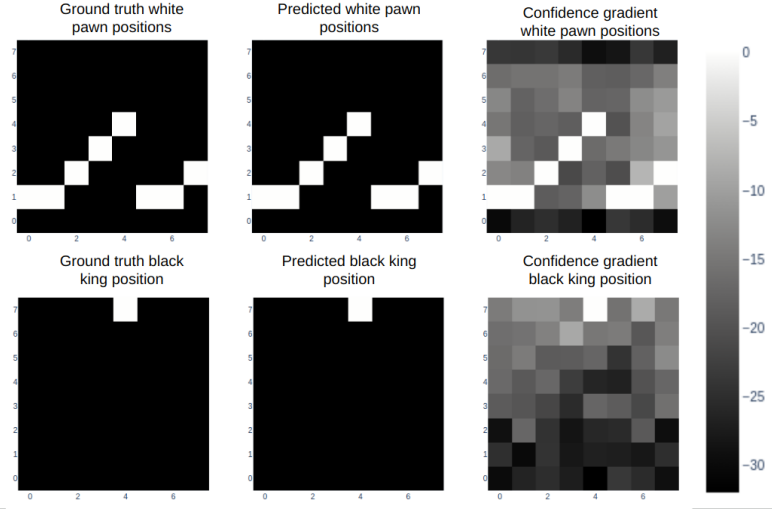


Figure 1.3: Probing analysis of a large language model trained to play chess [Karvonen, 2024].

191 where:

- 192 •  $h^l$  is the original LR of the input at layer  $l$ ,
- 193 •  $h^{l'}$  is the modified LR being optimized,
- 194 •  $h^{l*}$  is the optimal modified representation,
- 195 •  $\mathcal{L}_{\text{flip}}$ : loss encouraging the prediction of the target concept  $c_j$  to change
- 196 state (i.e., flipping the predicted label),
- 197 •  $\mathcal{L}_{\text{preserve}}$ : loss enforcing invariance of all non-target concepts  $c_k$  for  $k \neq j$ ,
- 198 •  $\|h^{l'} - h^l\|_2^2$ : regularization term ensuring the modified representation
- 199 remains close to the original LR,
- 200 •  $\gamma, \lambda, \beta$ : weighting coefficients controlling the trade-off between the inter-
- 201 vention strength, the preservation of other concepts, and the minimality of
- 202 the perturbation.

203 In the context of LLM studied previously, such interventions have been used  
 204 to remove specific pieces from the model’s internal representation of the game  
 205 state [Li et al., 2024, Karvonen, 2024]. The main observation is that, in the vast  
 206 majority of cases, the model behaves as if the piece had effectively disappeared.  
 207 The LLM continues to generate legal moves while correctly accounting for the  
 208 absence of the removed piece. This provides evidence for the causal role of these  
 209 internal representations in the model’s behavior.

210 A main limitation of the probing approach is that the concepts under in-  
 211 vestigation must be predefined before any analysis can be performed. This  
 212 introduces a strong bias in what can be studied. Moreover, annotating a dataset  
 213 for each concept is extremely time-consuming and requires significant human  
 214 effort. Interventions in LS are also difficult to interpret, as it is not always clear



what constitutes a meaningful change in the network’s output under a given perturbation.

Moreover, probing methods provide a binary view of concepts, focusing on whether a given concept is present or absent in a LR. This binary perspective can be overly restrictive, as concepts may be represented in a more continuous or graded manner, with varying degrees of presence. To obtain a more detailed characterization of these representations, other approaches have been proposed, such as the analysis of interpretable directions in LS.

### 1.3 Interpretable Directions

To overcome the limitations of binary concept detection in probing methods, several approaches have been proposed to analyze the structure of LR. Among them, the notion of interpretable directions has emerged as a natural extension, aiming to capture the continuous nature of how concepts are encoded in representation spaces. Empirically, it has been observed that shifting a LR along a direction can induce coherent and semantically meaningful changes in the model’s output [Haas et al., 2024, Voynov and Babenko, 2020]. Such transformations can be mathematically express as:

$$h' = h + \alpha \mathbf{v}_i,$$

where:

- $h$  denotes the original embedding,
- $\mathbf{v}_i$  is a unit vector associated with concept  $i$ ,
- $\alpha$  controls the strength of the intervention,
- $h'$  is the resulting perturbed representation.

In many cases, the resulting change in the model’s output varies smoothly with  $\alpha$ , suggesting an approximately linear CS with respect to certain semantic factors. This observation has led to the hypothesis that neural representations may organize concepts in a partially linear manner within the embedding space.

Vectors of this form are referred to as interpretable directions, as they correspond to transformations that can be associated with specific semantic attributes. The objective of this line of work is therefore to identify directions in LS that align with meaningful variations in the encoded concepts. This is illustrated schematically in Figure 1.4, where arrows represent semantic shifts from one concept toward others, and moving along these directions induces a coherent change in the model’s behavior, suggesting that the transition between concepts is continuously encoded along these axes.

This phenomenon has been extensively studied in generative image models. As illustrated in Figure 1.5, where each row corresponds to a distinct direction  $\mathbf{v}_i$  and each column to an increasing value of  $\alpha$ , controlled perturbations of LR along interpretable directions can induce specific semantic transformations, such as stroke thickness modification in digit images, head rotation in facial images, or texture and background changes in natural images [Voynov and Babenko, 2020]. These observations have led to the development of the field of LS editing, whose objective is to manipulate generated content directly through modifications in LS rather than through changes in the input prompt.

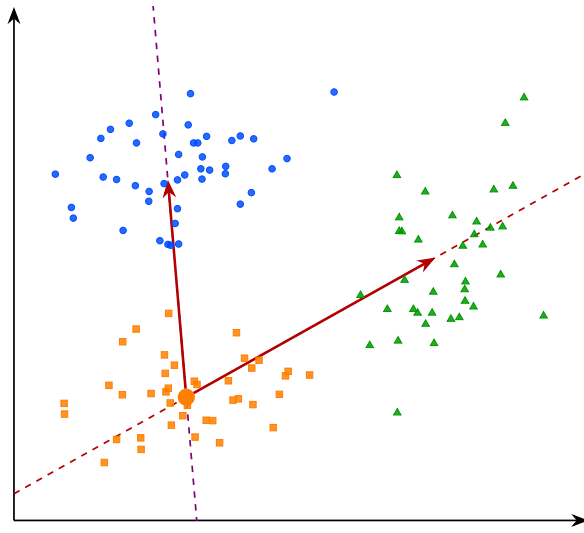


Figure 1.4: Interpretable directions in the latent space.

A central challenge remains identifying such interpretable directions. Most existing approaches rely on auxiliary models that encode human defined semantics, such as CLIP [Haas et al., 2024], to relate latent variations to textual concepts. Consequently, these methods inherit a limitation similar to that of probing techniques, since concept discovery depends on a pre-existing semantic framework defined by humans.

To overcome this limitation, a research direction known as unsupervised semantic discovery has emerged. Its goal is to uncover interpretable directions in LS without explicit semantic supervision. In these approaches, one model applies perturbations along random directions in LS, while a second model attempts to recognize the applied transformation. If a stable correspondence emerges between the perturbation and its prediction, this suggests that the considered direction corresponds to a coherent CS [Voynov and Babenko, 2020]. In this framework, human intervention occurs only a posteriori, to verify whether the discovered directions indeed correspond to interpretable semantic variations.

Despite these advances, several challenges remain. It appears that not all concepts encoded by NN are linearly identifiable in LS. In practice, interpretable direction methods typically discover only a few dozen directions per dataset, which intuitively seems insufficient to account for the full diversity of semantic variations that a generative model is capable of producing.

In addition, LR often exhibit a phenomenon known as entanglement, illustrated in Figure 1.6, where multiple semantic attributes are mixed within the same directions of the representation space. For instance, in a generative face model, moving along the “glasse” direction may simultaneously modify age and gender, whereas a disentangled representation would let each axis control a single semantic attribute while leaving the others unchanged. Ideally, one would like each semantic factor to vary independently along a specific direction, without affecting other attributes.

In practice, this is rarely the case, as a single direction in LS often encodes multiple semantic concepts simultaneously, such that modifying one concept



Figure 1.5: Interpretable directions in the latent space of a generative model [Voynov and Babenko, 2020].

can unintentionally affect several others. This lack of separability indicates that semantic information is not cleanly factored in the representation space, which motivates the search for more structured decompositions. This limitation has motivated a stronger hypothesis about how NN organize information internally, namely the *Superposition Hypothesis* (Hyp. 2), introduced in Section 1.4. Under this hypothesis, Sparse Autoencoder (SAE) provide a means of analyzing LR.

## 1.4 Sparse Autoencoder

Approaches based on latent directions face limitations. Only a restricted subset of concepts admit a linear representation, and even these directions are often entangled with multiple semantic concepts, as illustrated in Figure 1.6. To address this limitation, an extension of the *Concept Hypothesis* (Hyp. 1) has been proposed under the name *Superposition Hypothesis* (Hyp. 2), formalized in Hypothesis 2 [Elhage et al., 2022].

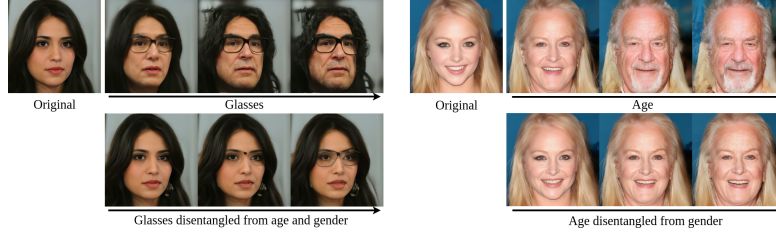


Figure 1.6: Illustration of entanglement in the latent space of a generative face model [Haas et al., 2024].

### Hypothesis 2: Superposition Hypothesis

Concepts are preferentially represented by distinct linear directions in LS. When the number of concepts exceeds the available dimensions, multiple concepts are encoded in superposition within the same directions.

According to this hypothesis, in an unconstrained LS, each concept is represented in its own independent linear direction. However, NN operate in finite-dimensional spaces, so when the number of concepts exceeds the available dimensions, such a one-to-one correspondence becomes impractical. The network must therefore reuse its representational dimensions, allowing multiple concepts to be encoded using the same directions. This shared encoding is referred to as superposition [Bricken et al., 2023]. As a result, in a superposed representation, linear analysis of the LS may not isolate individual concepts, the resulting directions reflect a mixture of several semantic attributes.

The idealized non-superposed representation and can be mathematically expressed as follows:

$$\mathbf{h} = \sum_{i=1}^n \alpha_i \mathbf{v}_i, \quad \forall i \neq j, \quad \mathbf{v}_i^\top \mathbf{v}_j \approx 0, \quad \|\boldsymbol{\alpha}\|_0 \ll n,$$

where:

- $\mathbf{h}$  denotes a LR,
- $n$  is the total number of concepts,
- $\mathbf{v}_i$  represents the direction associated with concept  $c_i$ ,
- $\alpha_i$  corresponds to the activation strength of concept  $c_i$  in the representation  $\mathbf{h}$ ,
- $\mathbf{v}_i^\top \mathbf{v}_j \approx 0$  indicates that concept directions are approximately orthogonal, meaning each concept independently controls a single factor of variation,
- $\|\boldsymbol{\alpha}\|_0 \ll n$  indicates that only a small fraction of concepts are active for a given representation.

Under the *Superposition Hypothesis* (Hyp. 2), the phenomenon of superposition becomes an obstacle for any method that attempts to analyze LR directly

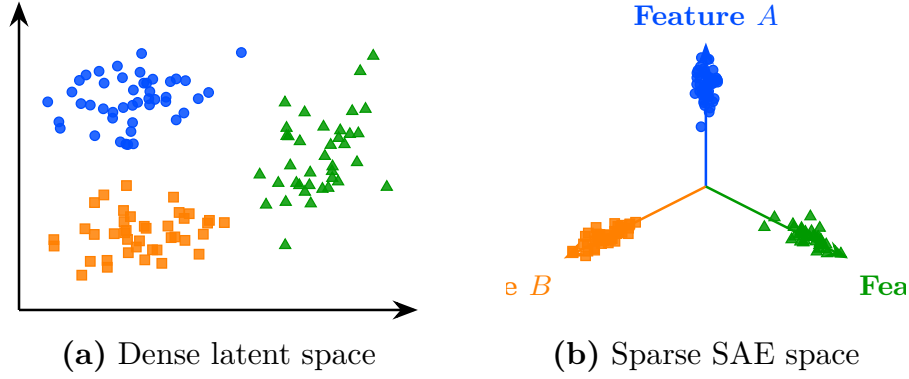


Figure 1.7: Transformation from a dense latent space (a) to a sparse SAE representation (b).

in the original activation space. This has motivated the development of methods that aim to recover a less superposed LR. A common approach is to map LS into a higher-dimensional space while imposing additional constraints that encourage individual concepts to be represented separately. This transformation is illustrated in Figure 1.7, while embeddings are entangled in the initial LS, the sparse representation aligns each concept with a dedicated feature axis.

SAE provide a practical implementation of this approach and can be defined as follows:

$$z = f_{\text{enc}}(h), \quad \hat{h} = f_{\text{dec}}(z)$$

$$\mathcal{L}_{\text{auto}} = \gamma \|h - \hat{h}\|_2^2 + \lambda \|z\|_1$$

where:

- $f_{\text{enc}}(h)$  is the encoding function that projects the original representation  $h$  into a higher-dimensional sparse space,
- $f_{\text{dec}}(z)$  is the decoding function that reconstructs the original representation from the sparse code,
- $z \in \mathcal{Z}$  is the sparse representation of  $h$ , with  $\dim(z) \gg \dim(h)$ ,
- $\hat{h}$  is the reconstructed representation,
- $\gamma \|h - \hat{h}\|_2^2$  is the reconstruction loss ensuring that the original representation can be faithfully recovered,
- $\lambda \|z\|_1$  is the sparsity regularization term that encourages most components of  $z$  to be zero,
- $\gamma, \lambda$  are weighting coefficients controlling the trade-off between reconstruction fidelity and sparsity.

In practice, both  $f_{\text{enc}}$  and  $f_{\text{dec}}$  are implemented as linear projections between the original representation space and the high-dimensional space  $\mathcal{Z}$ . This design follows the same principle as in probing methods, the transformation used to analyze the representation should remain as simple as possible. More expressive



Figure 1.8: A concept direction corresponding to the Golden Gate Bridge, discovered by a sparse autoencoder applied to a large language model [Templeton et al., 2024].

projection functions could themselves introduce complex structures into the transformed representation. In that case, it would become difficult to determine whether the observed concept reflects a CS already present in the model’s or structure introduced by the projection.

The parameters of these projection functions  $f_{\text{enc}}$  and  $f_{\text{dec}}$  are learned by minimizing  $\mathcal{L}_{\text{auto}}$  using gradient-based optimization, as introduced in Section ?? . Under this formulation, each dimension of  $\mathcal{Z}$  is interpreted as a dedicated concept direction.

The method proves to be powerful when successfully applied to language models [Templeton et al., 2024]. In these applications, SAE are used to desuperpose the LR of LLM. Once this representation is obtained, it has been shown that many concept vectors exhibit semantic properties. A example is the Golden Gate Bridge concept direction, illustrated in Figure 1.8, which activates systematically whenever the input text or image refers to the Golden Gate Bridge, regardless of the language or modality. This demonstrates that the learned concept is not tied to a specific surface form but captures a deeper semantic abstraction shared across modalities.

SAE also provide a way to intervene directly on the LR learned by the model. A concept can be artificially activated by modifying its corresponding feature in the sparse representation  $z$ , even when the input contains no reference to it. The modified representation is then projected back into the original activation space, after which the model proceeds with standard next-token generation from  $h^*$ :

$$h^* = f_{\text{dec}}(z^*), \quad \text{where} \quad z^* = z + \delta e_i,$$

where:

- $z$  is the original sparse representation of the input,
- $z^*$  is the modified sparse representation,
- $e_i$  is the basis vector of  $\mathcal{Z}$  associated with the target concept direction,
- $\delta$  controls the strength of the artificial activation,
- $h^*$  is the resulting edited representation in the original activation space.

With this method, a reference to the Golden Gate Bridge can be injected into the LLM’s internal representation. This causes the model’s output to shift

toward discussing the Golden Gate Bridge, even when the prompt contains no reference to it. This provides evidence that the learned feature plays a causal role in the model’s behavior. Such interventions enable control over specific model behaviors through LS editing.

Despite their empirical success, *Superposition Hypothesis* (Hyp. 2) exhibit several important limitations that question their assumptions. These limitations can be summarized as follows:

- **Instability of the decomposition.** Repeating the SAE training process multiple times on the same model does not yield the same decomposition of the LS. Different runs produce different concept directions in  $\mathcal{Z}$ , even when achieving comparable reconstruction performance, which raises questions about the reliability of the discovered CS.
- **Lack of semantic coverage.** In the high-dimensional space  $\mathcal{Z}$ , a large fraction of dimensions do not correspond to any interpretable semantic concept. This abundance of non-semantic directions raises questions about whether the decomposition truly reflects the model’s CS.
- **Sparsity–semantics trade-off.** Increasing the sparsity coefficient  $\lambda$  beyond a certain threshold degrades the semantic quality of the learned features [Jermyn et al., 2024]. Suggesting that enforcing excessive sparsity comes at the cost of interpretability.
- **Failure of scaling to remove superposition.** Under the *Superposition Hypothesis* (Hyp. 2), increasing the dimensionality of the representation space should reduce superposition. However, this is not observed in practice. Even in very high-dimensional LS, representations remain superposed, indicating that scaling alone does not resolve the entanglement.

These limitations suggest that *Superposition Hypothesis* (Hyp. 2), while powerful, do not fully resolve the problem of CS. Convincing empirical results do not necessarily validate the underlying theoretical assumptions.

Probing, interpretable directions, and *Superposition Hypothesis* (Hyp. 2) all rest on a shared assumption that the CS can be recovered through linear decompositions of LR. This assumption has driven significant empirical progress, yet it remains theoretically fragile. The following section therefore introduces a approach that relaxes this constraint.

## 1.5 Clustering

All the LS analysis methods presented so far rely on assumptions about a linear CS. For probing methods, concepts must be at least partially linearly separable for simple probes to succeed. Latent directions assumes that variations associated with a concept can be captured by moving along a linear axis of the LS. The *Superposition Hypothesis* (Hyp. 2) assumes that concepts are represented through linear directions that become entangled because of representational capacity constraints. Although these assumptions have led to empirical successes, their theoretical justification remains limited. Several observations suggest that the linearity hypothesis may not fully capture the CS.

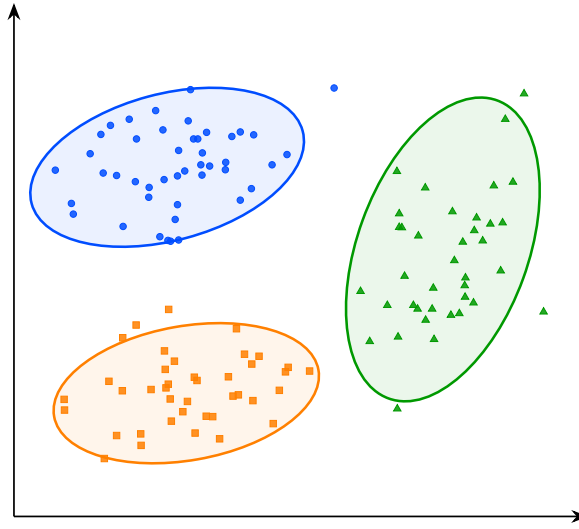


Figure 1.9: Visualization of the latent space structure.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity often degrades reconstruction quality and interpretability, suggesting that the underlying representation may not be perfectly described by a sparse linear decomposition. Taken together, these observations indicate that semantic concepts may not always exhibit a linear CS. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of LS.

To the best of our knowledge, clustering-based analysis is the only family of LS analysis methods that does not rely on any assumption of a linear CS [Ren et al., , Wei et al., ]. Rather than assuming that concepts correspond to directions, clustering methods only assume that conceptual similar inputs produce nearby LR. Conversely, inputs that differ significantly from a concept perspective are expected to be located farther apart in the LS. This is a considerably weaker assumption, in the sense that it is empirically verified in a wide range of settings.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of embeddings that occupy the same region of the LS. Concepts are therefore no longer associated with directions, but with clusters of neighboring representations. This is shown schematically in Figure 1.9, where the points represent embeddings associated with concepts A, B, and C, and the ellipses indicate the approximate extent of each group and highlight their geometric separation.

This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is projected into a LS using a NN. The resulting embeddings form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct concepts.

Deep clustering has been particularly successful in unsupervised image classification. Typically, an autoencoder is composed of an encoder  $f_{\text{enc}}$  and a decoder



454  $f_{\text{dec}}$ . The encoder maps an input  $x^{(i)} \in \mathcal{X}$  to a low-dimensional LR  $z^{(i)} \in \mathbb{R}^d$ :

$$z^{(i)} = f_{\text{enc}}(x^{(i)}), \quad d \ll \dim(\mathcal{X}).$$

455 To ensure that this compressed representation does not discard essential  
456 information, the decoder reconstructs the input from the LS:

$$\hat{x}^{(i)} = f_{\text{dec}}(z^{(i)}) = f_{\text{dec}}(f_{\text{enc}}(x^{(i)})).$$

457 The model is trained to minimize the reconstruction error between the original  
458 input and its reconstruction, typically using a Mean Squared Error (MSE) loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \|x^{(i)} - \hat{x}^{(i)}\|_2^2.$$

459 This objective encourages the LS to preserve the most informative features  
460 of the input while enforcing a compact representation.

461 Let  $\mathcal{D} = \{x^{(i)}\}_{i=1}^N$  denote a dataset of inputs. Through the encoder  $f_{\text{enc}}$ , this  
462 dataset induces a point cloud in the LS:

$$\mathcal{Z} = \{z^{(i)} \in \mathbb{R}^d \mid z^{(i)} = f_{\text{enc}}(x^{(i)}), x^{(i)} \in \mathcal{D}\}.$$

463 Clustering algorithms are then applied to this set  $\mathcal{Z}$  of embeddings in order  
464 to identify groups of similar representations.

465 An example of this methodology can be observed in Figure 1.10, where  
466 the approach is applied to the STL-10 dataset [Coates et al., 2011] using  $k$ -  
467 means clustering [MacQueen, 1967] with a number of clusters arbitrarily fixed  
468 to  $K = 10$  [Xie et al., ]. For each cluster, the corresponding line shows the 10  
469 samples closest to the associated centroid in the LS.

470 We observe that the method captures coherent semantic categories such  
471 as animals, vehicles, and flying entities. This result is particularly striking  
472 given that the only available information to the system consists of raw images.  
473 It suggests that the compression process induces a LS in which semantically  
474 meaningful CS emerge.

475 More advanced approaches combine representation learning and clustering  
476 within a single optimization process. In this setting, additional loss terms  
477 encourage embeddings to organize themselves around cluster centroids during  
478 training. This often produces LS that are easier to analyze and improves the  
479 quality of the discovered semantic CS [Guo et al., , Miklautz et al., ].

480 Despite its effectiveness, clustering-based analysis presents several important  
481 limitations. First, as with most unsupervised semantic discovery techniques,  
482 interpreting the resulting clusters remains partly subjective. Benchmark datasets  
483 often provide class labels for evaluation, but the discovered clusters may not  
484 match the categories that humans consider relevant. Alternative semantic  
485 groupings may also be equally valid. Second, clustering methods are highly  
486 sensitive to design choices, as the number of clusters, the distance metric, and  
487 the clustering algorithm itself can all significantly influence the results, making  
488 parameter selection challenging and often requiring substantial empirical tuning.  
489 Finally, to the best of our knowledge, no intervention technique has yet been  
490 proposed in the context of clustering-based analysis. Unlike probing or SAE  
491 analysis, which allow targeted modifications of LR to study their causal role,  
492 clustering methods currently offer no mechanism to alter the model’s behavior  
493 by acting directly on the discovered CS.

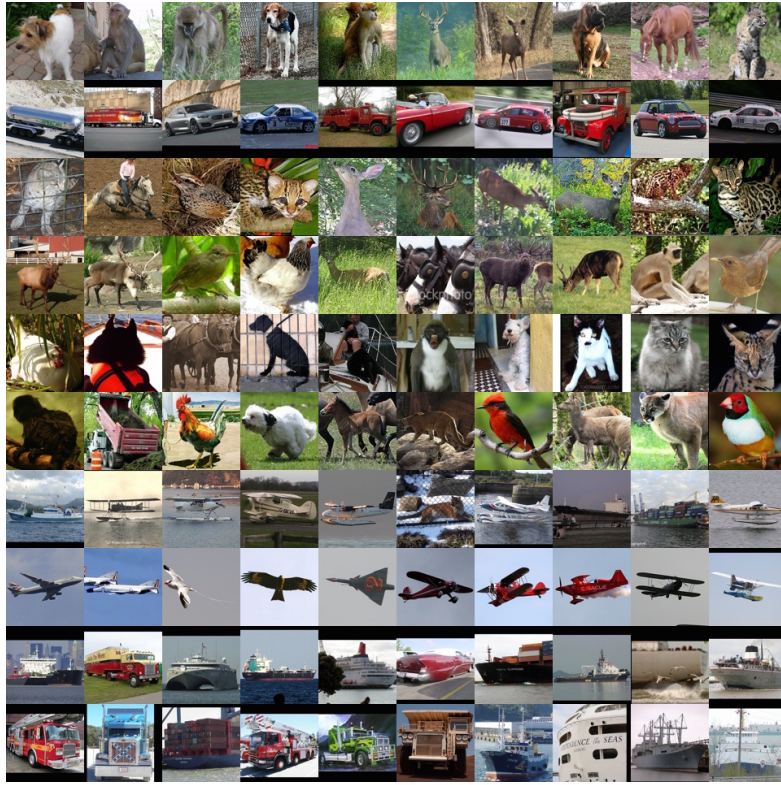


Figure 1.10: Clusters discovered by a deep clustering model applied to natural images [Xie et al., ].

## 1.6 Conclusion

The four families of methods reviewed in this chapter, namely probing, interpretable directions, sparse autoencoders, and clustering, are summarized in Figure 1.11 and all build upon the *Concept Hypothesis* (Hyp. 1) introduced in Section 1.1. Each of them approaches the question from a different angle, but they share a common objective: uncover how LR are structured and what information can be extracted from them.

However, a fundamental difficulty remains. There is currently no consensus on how to rigorously evaluate to what extent these methods genuinely provide access to the internal concepts of a model. Most existing metrics assess reconstruction quality or linear separability, but these do not directly measure whether the discovered CS correspond to meaningful abstractions from the model’s perspective. This leaves open the question of what it means, formally, for a concept to be represented in a NN.

Addressing this question may require contributions from disciplines beyond computer science and mathematics. As the notion of semantics is inherently tied to human interpretation, a purely formal or computational account may be insufficient. Insights from philosophy, cognitive science, or psychology could prove valuable in grounding the notion of concept in a way that is both theoretically rigorous and empirically tractable. Despite these open questions, the methods

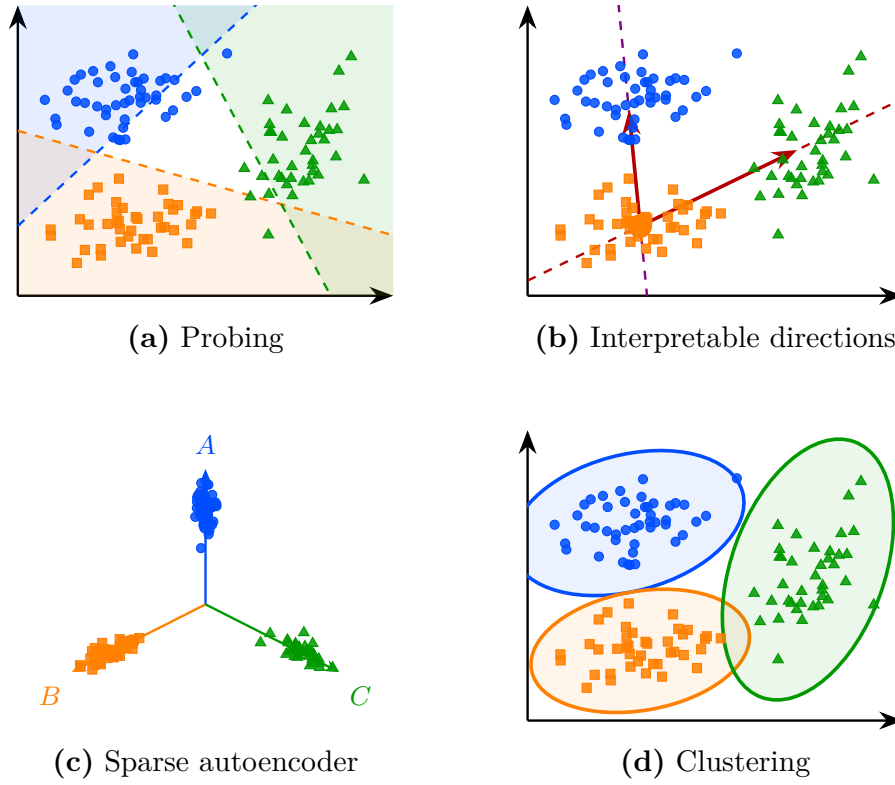


Figure 1.11: Overview of the four latent space analysis methods studied in this chapter, applied to the same point cloud.

514 studied in this chapter provide a practical foundation for analyzing the internal  
 515 representations of NN.

# 516 Acronyms

517 **CS** Concept Structure 4, 8, 9, 13–17

518 **DL** Deep Learning 1

519 **LLM** Large Language Model 1, 5, 7, 13

520 **LR** Latent Representation 1, 2, 5–17

521 **LS** Latent Space 1–5, 7–9, 11, 12, 14–16

522 **MSE** Mean Squared Error 16

523 **NN** Neural Network 1, 2, 4, 5, 9–11, 15, 17, 18

524 **RL** Reinforcement Learning 1

525 **SAE** Sparse Autoencoder 10, 12–14, 16

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